

Karan Kumar

📍 Sukkur, Pakistan | 📞 +92-312-3269988 | ✉ karantejwani48@gmail.com | 🌐 karantejwani | 🗣 KaranTejwani | 🌐 Portfolio

EDUCATION

Sukkur IBA University

Bachelor of Science in Computer Science | CGPA: 3.73

- Scholarship: Sindh Talent Hunt Program a merit-cum-need based scholarship
- Relevant Coursework: Deep Learning, Machine Learning, Computer Vision, Data Structures & Algorithms, Software Engineering, Database Systems, Computer Networks

Sukkur, Pakistan

2022 – 2026

EXPERIENCE

DeepNLP.ai

AI Research Collaborator

- Designed and executed deep learning experiments on Transformer and CNN-based architectures for NLP and computer vision research tasks
- Co-authored two IEEE-accepted publications; contributed to experiment design, quantitative analysis, and technical writing
- Managed independent research pipelines end-to-end with minimal supervision

Remote

Oct 2025 – Present

National Assembly of Pakistan

AI Intern

- Developed and deployed a production-grade bilingual (Urdu/English) transcription and translation system using OpenAI Whisper for institutional parliamentary use
- Delivered full-stack solution independently: Flask REST API backend, React frontend, deployed on Linux infrastructure
- System actively used by parliamentary staff for daily documentation workflows

Islamabad, Pakistan

Jun 2025 – Aug 2025

Simper AI - NEP NIC Sukkur

Co-Founder

- Co-built an NLP-powered EdTech platform automating exam generation
- Selected and incubated through NEP NIC Sukkur, a nationally recognized startup incubation program

Sukkur, Pakistan

May 2024 – Aug 2024

PUBLICATIONS

First Author | *Analyzing the Classification-Localization Gap Between CNNs and Vision Transformers*

IEEE iCoMET 2026

Co-Author | *Leveraging Transformers & XAI to Identify Helpful Student Reviews*

IEEE EDUCON 2026

PROJECTS

NeuroEmotion | GitHub

- Benchmarked **18 deep learning models** (CNNs, LSTMs, GRUs, Transformers) across DREAMER and SEED-IV EEG datasets with multiple preprocessing pipelines
- Selected LSTM for production based on full TFLite compatibility; deployed as Flutter mobile app with real-time emotion classification via Flask REST API

Final Year Project

Image Manipulation Detection and Segmentation | GitHub

- Trained and evaluated image manipulation detection models on the DeFacto dataset, supporting classification of splicing, copy-move, inpainting, and face-manipulation forgeries.
- Built an end-to-end computer vision platform with pixel-level forgery localization, exposing inference through scalable Flask REST APIs and interactive Next.js frontends.

Computer Vision

Healr - Healthcare Platform | GitHub

- Engineered a MERN-stack healthcare platform serving patients, doctors, and administrators, supporting appointment booking, medicine search, lab test management, and role-based access control.
- Integrated external healthcare APIs, implemented secure authentication and validation using Formik/Yup, and deployed the application on Vercel with a scalable Node.js/MongoDB backend.

Full-Stack

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, Hugging Face, scikit-learn, OpenCV, LangChain, TFLite, ONNX

Architectures: CNNs, Vision Transformers, LSTM, GRU, YOLO, Faster R-CNN, RT-DETR, LLMs, RAG

Engineering: Python, Flask, REST APIs, React.js, Node.js, Flutter, Docker, Linux, Git

Concepts: MLOps, XAI, Signal Processing, NLP, System Design, SDLC, OOP

LEADERSHIP & COMMUNITY

Programming Resource Centre Lead | CS Society, Sukkur IBA University

- Led technical workshops, coding sessions, and peer mentorship programs for freshers in university

Sep 2024 – May 2025

Board Member | Sindh Educational Organization

- Managed and Hosted 50+ virtual sessions on career counseling and education with local and international speakers

Feb 2024 – May 2026